

Unit 4 Knowledge and Reasoning

Introduction

☐ We have looked at different ways to represent knowledge in computers.

☐ Even though these methods have some benefits, their meanings are not always very clear.

Example:

If we store knowledge like “*Birds can fly*” as a simple rule, it may not explain exceptions like penguins.

- **Logic** is a way of representing knowledge that *does* have clear and precise meaning.
- That’s why logic is often used in **ontologies**

Example:

In logic, we can write:

“If something is a bird AND it is not a penguin, THEN it can fly.”

This removes confusion.

Knowledge-Based Agents

☐ A **knowledge base** is a collection of sentences written in a formal language that a computer can understand.

☐ In the **declarative approach**, we build an intelligent agent by simply **telling it facts and rules** instead of programming every action.

- We tell the agent **what it needs to know**.
- Then the agent can **decide by itself what to do**, and the answers come from the knowledge stored in the knowledge base.

☐ We can look at agents at the **knowledge level**.

- This means we focus on **what the agent knows**, not on **how it is programmed or implemented internally**.

A Practical Application of Knowledge Representation (KR)

1. What is Knowledge Representation?

Knowledge Representation is a way of storing information in a computer so that it can **understand, reason, and make decisions** like a human.

It answers questions such as:

- What does the system know?
- How can it use that knowledge to solve problems?

2. Why Knowledge Representation is Important

- Helps machines **store real-world knowledge**
- Allows systems to **reason and make decisions**
- Makes AI systems **intelligent and flexible**
- Used in many real-life applications

3. Practical Applications of Knowledge Representation

1. Expert Systems

Expert systems imitate the decision-making ability of human experts.

Example: Medical Diagnosis System

- Knowledge stored:
 - “If a patient has fever and cough → possible flu”
- The system uses this knowledge to suggest a diagnosis.

Use:

- Hospitals
- Engineering troubleshooting
- Financial advice

2. Search Engines

Search engines use knowledge representation to understand user queries.

Example: Google Search

- Understands relationships between words
- Knows that “capital of India” → “New Delhi”

Use:

- Better search results
 - Question answering
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3. Robotics

Robots use knowledge representation to understand their environment.

Example: Cleaning Robot

- Knowledge:
 - “If obstacle ahead → turn left”
 - Helps the robot move safely and efficiently.
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4. Natural Language Processing (NLP)

KR helps machines understand human language.

Example: Chatbots

- Understand user input
- Respond using stored knowledge

Use:

- Customer support

- Virtual assistants (Siri, Alexa)
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5. Autonomous Vehicles

Self-driving cars use KR to make driving decisions.

Example:

- “If traffic light is red → stop”
 - “If pedestrian detected → slow down”
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6. Medical Information Systems

Used to store and reason about patient data.

Example:

- Symptoms + test results → treatment suggestion
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4. Types of Knowledge Used in Applications

- **Declarative knowledge** – facts (e.g., “Paris is the capital of France”)
 - **Procedural knowledge** – how to do things (e.g., steps to diagnose a disease)
 - **Heuristic knowledge** – experience-based rules
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5. Advantages of Knowledge Representation

- Easy to update knowledge
- Improves decision-making
- Makes systems more intelligent

Overview of Logic in Artificial Intelligence (AI)

What is Logic in AI?

Logic in AI is a **formal way to represent knowledge and reason about it**. It helps AI systems **draw correct conclusions** from given facts.

Why Logic is Used in AI

- Provides **clear and precise meaning**
- Supports **reasoning and decision making**
- Widely used in **knowledge representation**
- Helps AI systems act **rationally**

Applications of Logic in AI

- Expert systems
- Automated reasoning
- Knowledge-based agents
- Ontologies (e.g., Semantic Web)

Types of Logic in AI

- Propositional Logic
- First-Order Logic
- Modal Logic
- Description Logic

Propositional Logic

- Propositional Logic is the **simplest form of logic** used in AI
It deals with **statements (propositions)** that are either **true or false**
- Describe what we know about a particular domain by a propositional formula, KB

Syntax of Propositional Logic

Syntax defines the **structure and rules** for writing logical sentences.

Components of Syntax

1. Propositional Symbols

- Represent simple statements
- Usually written as:
P, Q, R, A, B

Example:

P: "It is raining"

Q: "The ground is wet"

2. Logical Connectives

Symbol	Name	Meaning	Example
$\neg$	NOT	Negation	$\neg P$
$\wedge$	AND	Conjunction	$P \wedge Q$
$\vee$	OR	Disjunction	$P \vee Q$
$\rightarrow$	IMPLIES	If-then	$P \rightarrow Q$
$\leftrightarrow$	IF AND ONLY IF	Biconditional	$P \leftrightarrow Q$

3. Well-Formed Formulas (WFF)

A sentence is **well-formed** if it follows syntax rules.

Example of WFF:

- $(P \wedge Q)$
- $(P \rightarrow Q)$

Not a WFF:

- $\wedge P Q$
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4. Semantics of Propositional Logic

Semantics defines the **meaning** of logical sentences.

It tells us:

- When a sentence is **true or false**
 - How truth values are assigned
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Truth Values

- **True (T)**
 - **False (F)**
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Interpretation / Model

An interpretation assigns **truth values** to propositional symbols.

Example:

$P = \text{True}$

$Q = \text{False}$

Truth Tables (Meaning of Connectives)

NOT ($\neg$)

P $\neg$ P

T F

F T

AND ($\wedge$)

P Q P $\wedge$ Q

T T T

T F F

F T F

F F F

OR ($\vee$)

P Q P $\vee$ Q

T T T

T F T

F T T

F F F

IMPLIES ($\rightarrow$)

P Q $P \rightarrow Q$

T T T

T F F

F T T

F F T

Validity and Satisfiability

- **Satisfiable:** True in at least one interpretation
 - **Unsatisfiable:** False in all interpretations
 - **Valid (Tautology):** True in all interpretations
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5. Example

Let:

- P = "It is raining"
- Q = "Road is wet"

Sentence:

$P \rightarrow Q$

Meaning:

"If it is raining, then the road is wet."

6. Importance of Propositional Logic in AI

- Simple and easy to understand
- Foundation for advanced logics
- Used in rule-based systems

Propositional Theorem Proving

What is Propositional Theorem Proving?

Propositional Theorem Proving **is the process of** checking whether a conclusion logically follows from a given set of propositions (knowledge base) **using rules of logic.**

Basic Terms:

1. Knowledge Base (KB)

A set of propositional logic sentences assumed to be true.

2. Query

A statement we want to prove from the KB.

3. Entailment ($\models$)

$KB \models \alpha$ means:

α is true in every model where KB is true.

α is true in every model where KB is true.

- Common Methods of Propositional Theorem Proving :

- *Truth Table Method*

Description

- Lists all possible truth values of propositions
- Checks if the query is true in all cases where KB is true

Example

KB:

1. $P \rightarrow Q$
2. P

Query: Q

P Q $P \rightarrow Q$ KB True? Q

T T T Yes T

T F F No F

F T T No T

F F T No F

✓ In all rows where KB is true, Q is true.

☞ **Q is proved**

Advantages

- Simple and accurate

Disadvantages

- Not efficient for many variables

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- Inference Rules (Logical Deduction)

Uses logical rules to derive new facts.

Common Inference Rules

Modus Ponens

If:

- $P \rightarrow Q$
 - P
- Then:
- Q
-

Example

KB:

1. If it rains, the ground is wet ($P \rightarrow Q$)
2. It is raining (P)

Conclusion:

✓ The ground is wet (Q)

- *Resolution Method*

What is Resolution?

A **rule-based proof method** used for automated theorem proving.

It works by:

- Converting all sentences into **Conjunctive Normal Form (CNF)**
 - Applying the **resolution rule**
 - Deriving a contradiction (empty clause)
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Resolution Rule

From:

- $(A \vee B)$
- $(\neg A \vee C)$

We get:

- $(B \vee C)$

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- Steps in Resolution Proof
 1. Convert KB and $\neg$ Query into CNF
 2. Add them together
 3. Apply resolution repeatedly
 4. If **empty clause ($\perp$)** is derived $\rightarrow$ query is proved
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Resolution Example

KB:

1. $P \rightarrow Q$
2. P

Query: Q

Step 1: Convert to CNF

- $P \rightarrow Q \Rightarrow \neg P \vee Q$
 - $P \Rightarrow P$
 - $\neg Q$ (negation of query)
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Step 2: Resolution

1. $(\neg P \vee Q)$ and $(\neg Q) \Rightarrow \neg P$
2. $(\neg P)$ and $(P) \Rightarrow \perp$ (empty clause)

✓ Empty clause found

☞ **Q is proved**

Example: Proving a Conclusion Using Propositional Logic

Knowledge Base (KB):

1. If a student studies, then the student will pass the exam.
 $\rightarrow P \rightarrow Q$
2. The student studies.
 $\rightarrow P$

Query (What we want to prove):

- The student will pass the exam.
 $\rightarrow Q$

Method 1: Using Modus Ponens

Given:

- $P \rightarrow Q$
- P

Apply Modus Ponens:

- Therefore, Q

✓ Conclusion is proved

Method 2 :Using Resolution

Step 1: Convert to CNF

- $P \rightarrow Q \Rightarrow \neg P \vee Q$
- P
- Negation of query: $\neg Q$

Step 2: Apply Resolution

1. $(\neg P \vee Q)$ and $(\neg Q) \Rightarrow \neg P$
2. $(\neg P)$ and $(P) \Rightarrow \perp$ (empty clause)

Result:

Since we got an empty clause, the query Q is proved.